

BIJAN MAZAHERI

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I am interested in the information theory of causality, experimental design, and artificial intelligence. I am motivated by applications in defense and biology.

EDUCATION

California Institute of Technology - Pasadena, CA <i>Doctorate of Philosophy</i> Computing and Mathematical Sciences, GPA: 3.9/4.0 Thesis Title: Combining Sources and Leveraging Contexts Advised by Prof. Leonard Schulman and Prof. Jehoshua (Shuki) Bruck	<i>Oct 2017 - Aug 2023</i>
Cambridge University - Cambridge, UK Classes in the Mathematical Tripos Parts IA, IB, and II. Supported by a Herchel Smith Fellowship.	<i>Oct 2016 - Jun 2017</i>
Williams College - Williamstown, MA <i>Bachelor of Arts</i> Physics and Computer Science, GPA: 3.92/4.00 Highest Honors (Physics), Phi Beta Kappa, Sigma Xi, Magna Cum Laude Thesis Title: RNA Macrostates and Macrokinetics	<i>Sep 2012 - Jun 2016</i>

ACADEMIC APPOINTMENTS

Thayer School of Engineering, Dartmouth College - Hanover, NH <i>Assistant Professor of Engineering</i> Affiliate - Quantitative and Biological Sciences Affiliate - Computational Science and Modeling Program	<i>Jan 2025 - Present</i>
Broad Institute of MIT and Harvard - Cambridge, MA <i>Eric and Wendy Schmidt Postdoctoral Associate</i> <i>Visiting Research Scientist</i> Primarily advised by Prof. Caroline Uhler.	<i>Oct 2023 - Dec 2024</i> <i>Jan 2025 - Present</i>

INDUSTRY EXPERIENCE

Operand – San Francisco, CA <i>Founding Research Advisor</i>	<i>Apr 2026 - Present</i>
Amazon Research Causality Lab - Tübingen, Germany <i>Intern</i> Worked with Dr. Michaela Hardt, Dr. Atalanti Mastakouri, and Dr. Dominik Janzing	<i>Oct 2022 - Feb 2023</i>
BioDiscovery - El Segundo, CA <i>Intern</i> Worked with Dr. Soheil Shams.	<i>Jun 2017 - Sep 2017</i>
IBM T.J. Watson Research Center - Yorktown Heights, NY <i>Intern</i>	<i>Jun 2016 - Sep 2016</i>

Worked with Dr. Victor Kravets (mentor) and Dr. Andrew Sullivan (manager).

AWARDS AND GRANTS

Dartmouth Initiative for Middle East Exchange (DIMEX) “AISTATS 2026 and Resulting Collaborations.” \$22,000.	<i>Awarded Fall 2025</i>
DARPA Advanced Research Concepts - COMPASS (Lead PI) “Detecting Data Manipulation via Causal Information Geometry.” \$248,000.	<i>Awarded Summer 2025</i>
Eric and Wendy Schmidt Postdoctoral Fellowship 1-3 years of research at the Broad Institute of MIT and Harvard. \$225,000 approx.	<i>Awarded Summer 2023</i>
Amazon AI4Science Research Fellowship “Expert Graphs and Source Identification.” \$25,000.	<i>Awarded Spring 2022</i>
National Science Foundation Graduate Research Fellowship 3 year Ph.D. fellowship. \$138,000 approx.	<i>Awarded Spring 2019</i>
Herchel Smith Fellowship Funding for 1-3 years of study at Cambridge University. \$100,000 approx.	<i>Awarded Spring 2016</i>

TEACHING

Dartmouth Engineering Discrete and Probabilistic Systems Principles of Causality	<i>Summer 2025</i> <i>Spring 2025, 2026</i>
Teaching Assistant Markov Chain Monte Carlo (Caltech) Discrete Mathematics (Williams) Electricity and Magnetism (Williams) Introduction to Mechanics (Williams)	<i>Spring 2022</i> <i>Fall 2015</i> <i>Spring 2015</i> <i>Fall 2013</i>

PUBLICATIONS

Bijan Mazaheri, Chandler Squires, Caroline Uhler. Synthetic Potential Outcomes and Causal Mixture Identifiability. In *The 28th International Conference on Artificial Intelligence and Statistics*, 2025.

Bijan Mazaheri, Siddharth Jain, Matthew Cook, Jehoshua Bruck. Omitted Labels Induce Nontransitive Paradoxes in Causality. In *4th Conference on Causal Learning and Reasoning*, 2025.

Spencer Gordon, Eric Jahn, ***Bijan Mazaheri**, Yuval Rabani, and Leonard J Schulman. Identification of Mixtures of Discrete Product Distributions in Near-Optimal Sample and Time Complexity. In *The Thirty Seventh Annual Conference on Learning Theory*, pp. 2071-2091. PMLR, 2024.

Bijan Mazaheri, Atalanti Mastakouri, Dominik Janzing, and Michaela Hardt. Causal Information Splitting: Engineering Proxy Features for Robustness to Distribution Shifts. In *The 39th Conference on Uncertainty in Artificial Intelligence*, 2023.

Spencer Gordon, ***Bijan Mazaheri**, Yuval Rabani, and Leonard J Schulman. Causal Inference Despite Limited Global Confounding via Mixture Models. In *2nd Conference on Causal Learning and Reasoning*, 2023.

Siddharth Jain, **Bijan Mazaheri**, Netanel Raviv, and Jehoshua Bruck. Glioblastoma signature in the DNA of blood-derived cells. *PLOS One* 16(9): e0256831. 2021.

Bijan Mazaheri, Siddharth Jain, and Jehoshua Bruck. Expert Graphs: Synthesizing New Expertise via Collaboration. In *2021 IEEE International Symposium on Information Theory (ISIT)*, pages 2447–2452, 2021.

Spencer Gordon, ***Bijan Mazaheri**, Yuval Rabani, and Leonard Schulman. Source Identification for Mixtures of Product Distributions. In *The Thirty Fourth Annual Conference on Learning Theory*, pages 2193–2216. PMLR, 2021.

Bijan Mazaheri, Siddharth Jain, and Jehoshua Bruck. Robust Correction of Sampling Bias using Cumulative Distribution Functions. *Advances in Neural Information Processing Systems*, volume 33, pages 3546–3556. Curran Associates, Inc., 2020.

* = Authorship order is alphabetical.

PREPRINTS

Zou Yang, Sophia Xiao, and **Bijan Mazaheri**. Masked Unfairness: Hiding Causality in Zero ATE. Under Review. 2026.

Liyuan Xu and **Bijan Mazaheri**. Estimating Aleatoric Uncertainty in the Causal Treatment Effect. Under Review. 2026.

Mateusz Gajewski, Sophia Xiao, and **Bijan Mazaheri**. Data Augmentation via Causal Residual Bootstrapping. Under Review. 2026.

Bijan Mazaheri, Jiaqi Zhang, and Caroline Uhler. Relaxing Faithfulness for Intervention-Only Causal Discovery. Under Review. 2025.

Bijan Mazaheri, Jiaqi Zhang, and Caroline Uhler. Meta-Dependence in Conditional Independence Testing. *arXiv:2504.12594*. Under review. 2025.

Bijan Mazaheri, Spencer Gordon, Yuval Rabani, and Leonard Schulman. Causal Discovery under Latent Class Confounding. *arXiv:2311.07454*, Under Review, 2025.

Spencer Gordon, ***Bijan Mazaheri**, Yuval Rabani, and Leonard J Schulman. The sparse Hausdorff moment problem, with application to topic models. *arXiv:2007.08101*, 2020.

Siddharth Jain, **Bijan Mazaheri**, Netanel Raviv, and Jehoshua Bruck. Cancer Classification from Healthy DNA using Machine Learning. *BioRxiv*, page 517839, 2019.

Siddharth Jain, **Bijan Mazaheri**, Netanel Raviv, and Jehoshua Bruck. Short Tandem Repeats Information in TCGA is Statistically Biased by Amplification. *BioRxiv*, page 518878, 2019.

* = Authorship order is alphabetical.

PATENTS

Siddharth Jain, **Bijan Mazaheri**, Netanel Raviv, and Jehoshua Bruck. Mutation profile and related labeled genomic components, methods and systems. 2019.

INVITED TALKS

CauScale Spotlight @ AISTATS Title: “Masked Unfairness: Hiding Causality in Zero ATE.”	<i>May 2026</i>
Tshinghua Sanya International Mathematics Forum Title: “Distribution-level Techniques for Heterogeneous Causal Relationships”	<i>Jan 2026</i>
Biomedical Data Science Seminar Series at Dartmouth Medical School Title: “Causal Underpinnings of Information Synthesis”	<i>Oct 2025</i>
Williams College Statistics Colloquium Title: “Causal Underpinnings of Information Synthesis”	<i>Oct 2025</i>
Ops/MS Brown Bag Seminar, Tuck School of Business Title: “Synthetic Potential Outcomes and Causal Mixture Identifiability”	<i>Feb 2025</i>
Boston University Machine Learning Symposium Title: “Synthetic Potential Outcomes and Causal Mixture Identifiability”	<i>Nov 2024</i>
Stanford Online Causal Inference Seminar Title: “Synthetic Potential Outcomes and the Hierarchy of Causal Identifiability”	<i>Oct 2024</i>
Jones Seminar, Thayer School of Engineering at Dartmouth Title: “Latency and Heterogeneity in Data and What to do About it”	<i>May 2024</i>
Simon’s Institute for Theory of Computing Title: “Causal Discovery under Limited Global Confounding”	<i>May 2023</i>

RESEARCH MENTORSHIP

PhD Advisees

Zou Yang (Thayer School of Engineering, Dartmouth)
Sophia Xiao (Thayer School of Engineering, Dartmouth)

Graduate Students

Mateusz Gajewski (PhD Student, Poznań University of Technology)
Ryan Montgomery (MBA Student, Tuck School of Business, Dartmouth)

Undergraduate

Youmi Ji (Dartmouth College URAD)
Keion Grieve (Dartmouth College URAD)
Caroline Krantz (Dartmouth College URAD)
Muhammad Ahmad (Dartmouth College FYREE)
Hailey King (Dartmouth College)
Ruchita Nair (Dartmouth College)
Benjamin Cavanagh (Dartmouth College)
Sreshth Tiwari (Dartmouth College URAD)
Connor Kilkenny (Dartmouth College URAD)
Titus Johnson (Dartmouth College URAD)
Tamier Baoyin (B.A. Mt. Holyoke College, B.E. Dartmouth College)
Leila Salken (Williams College)
Anika Roy (B.S. IIIT Hyderabad, Khorana Scholar, Heidelberg Future Leaders Award)
Beshr Bouli (MIT UROP)

Data Science Projects

I have supported over 30 projects with undergraduate students using data on my website, and have advised some of these students in applying to graduate school.

WORKSHOPS

CauScale @ AISTATS Title: Masked Unfairness: Hiding Causality in Zero ATE.	<i>May 2026</i>
Tsinghua Sanya International Mathematics Forum Title: Causality and Machine Learning.	<i>Jan 2026</i>
CauScien Workshop at NeurIPS Attendee.	<i>Dec 2025</i>
Scaling Up Intervention Models at ICML Title: Faithfulness and Intervention-Only Causal Discovery .	<i>Jul 2025</i>
Simon's Institute for Theory of Computing Title: Causality.	<i>Spring 2022</i>

SERVICE

Conferences

CLear: Reviewer (2024-2026), Program Chair (2026).
NeurIPS: Reviewer (2020 - 2025), Top Reviewer Award (2025)
UAI: Reviewer (2026)
AISTATS: Reviewer (2023 - 2026)
ICML: Reviewer (2025)

Journals

Nature Machine Intelligence: Reviewer (2024)
The American Statistician: Reviewer (2024)
Electronic Journal of Statistics: Reviewer (2025)

Committees

Dartmouth Engineering 3-2 and 2-1-1-1 Programs
Curriculum, Dartmouth Computational Science and Modeling Program

Other Service

Faculty Advisor, Dartmouth Cross Country and Track and Field Teams
Volunteer Assistant Coach, Caltech Cross Country and Track and Field Teams
Sports statistics outreach for Cross Country and a correspondent at D3 Glory Days

PROJECTS

LACCTiC I maintain a website for collegiate cross country with 10,000 regular users that applies concepts from batch-effect correction to ranking performances on differing terrain. The backend runs on Python and Django and the frontend uses React, and the database is hosted on AWS. I have helped advise over 30 student projects using this data.	<i>Sep 2021 - present</i>
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